JEREMY ZAY

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EDUCATION

***Wesleyan University*,** Middletown, CT
MA in Computer Science | Expected Grad: May 2027
BA in Computer Science (Honors), Mathematics, Integrative Sciences (Honors) | GPA: 3.74 | May 2026

EXPERIENCE

Software Engineer Intern, Chatham Financial | Denver, CO | Jun 2026 – Aug 2026
- Owned and built Curve Explorer within ONYX Pro in React and TypeScript, consolidating curve visualization, comparison, and rebuilding into a single workflow and eliminating 10+ manual steps.
- Integrated the frontend with C#/.NET REST APIs using Swagger/OpenAPI to fetch curve data.
- Wrote Vitest unit tests covering 90% of new code, meeting CI requirements for coverage, linting, SonarCloud analysis, and automated tests.

ML Research Fellow, Thayer Lab, Wesleyan University | Middletown, CT | May 2024 – Present
- Developed a PyTorch diffusion framework generating 4,000 molecules per experiment and evaluated 10,000+ generated structures using chemical validity and molecular docking.
- Designed reusable containerized training and evaluation pipelines for deep learning models on HPC clusters.
- Achieved 97% molecular validity and best docking score of -8.94 kcal/mol on p53.
- Awarded $25K+ in research funding from Wesleyan University across multiple fellowships and grants.

Founder & Software Engineer, Sonar (jeremyzay.com/sonar) | Mar 2026 – Present
- **F**ounded and launched Sonar, an iOS podcast tracking app built in Swift/SwiftUI and Supabase/PostgreSQL, growing to 40+ users with 570+ episode logs and 160+ reviews in its first 6 months.
- Built a ~55K-line codebase with 550+ unit tests, including ratings and reviews, social features, push notifications, Sign in with Apple and an AI recommendation guide.
- Lead product, branding and marketing; designed a Postgres backend secured with 58 row-level security policies and automated TestFlight and App Store releases with fastlane.

Scientific Computing and Informatics Center Tutor (SCIC), Wesleyan University | Sep 2025 – Present
- Provided 300+ hours of one-on-one technical consulting to 50+ students, graduate researchers, and faculty in scientific computing and software development.
- Debugged research code and reviewed student projects in HPC execution, parallel jobs, and software design.
- Taught AI (graph search, probabilistic reasoning and RL) to 200+ students as a TA across 6 CS courses.

PROJECTS

karchive (karchive.dad) | Aug 2026 – Present
- Built a searchable YouTube archive that lets users ask questions and get timestamp-cited answers across 1,000+ transcripts, using Gemini embeddings stored in Supabase/PostgreSQL and a React/FastAPI stack.
- Built a hybrid RAG pipeline over 4M+ embedded transcript tokens, retrieving ~6K tokens of relevant context per query to generate grounded answers with video timestamps.

Zaychess (jeremyzay.com/zaychess) | Jun 2025 – Mar 2026
- Shipped a Java chess application on the macOS App Store with on-device AI opponents.
- Built a Python matchmaking server that queues players, pairs opponents, and relays moves between clients.
- Trained a residual policy/value network through 500K self-play games and MCTS, improving evaluation strength by 600 Elo over the initial model while processing 3,000 tree nodes/sec.

RELEVANT COURSEWORK

Applications: Machine Learning, Artificial Intelligence, Program Analysis, Computer Networks
Theory: Real Analysis, Automata Theory, Algorithms & Complexity, Probability & Statistics, Linear Algebra

SKILLS

Languages: Python, TypeScript, Java, C#, C, Swift, SQL, Bash
Frameworks: React, FastAPI, Flask, .NET, PyTorch, scikit-learn, NumPy, pandas
Tools: Docker, GitHub Actions, CI/CD, Git, Jira, Linux, HPC